

Several Complex Variables

Labix

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1 Holomorphic and Analytic Functions

1.1 Polydiscs

Definition 1.1.1 (Polydiscs)

Let $p = (p_1, \dots, p_n) \in \mathbb{C}^n$. Let $r = (r_1, \dots, r_n) \in \mathbb{N}^n$. Define the polydisc of polyradii r at p to be

$$D_r(p) = \{(z_1, \dots, z_n) \in \mathbb{C}^n \mid |z_i - p_i| < r_i \text{ for } 1 \leq i \leq n\}$$

Definition 1.1.2 (Shilov Boundary)

Let $p = (p_1, \dots, p_n) \in \mathbb{C}^n$. Let $r = (r_1, \dots, r_n) \in \mathbb{N}^n$. Define the Shilov boundary of the polydisc $D_r(p)$ by

$$T_r(p) = \{(z_1, \dots, z_n) \in \mathbb{C}^n \mid |z_i - p_i| = r_i \text{ for } 1 \leq i \leq n\}$$

The Shilov boundary is not quite the topological boundary of the polydisc: it is strictly smaller.

1.2 Holomorphic and Analytic Functions

Definition 1.2.1 (Partial Derivatives)

Let $U \subseteq \mathbb{C}^n$ be open. Let $f = (f_1, \dots, f_n) : U \rightarrow \mathbb{C}$ be a function. For $1 \leq k \leq n$, define the partial derivatives

$$\frac{\partial f}{\partial z_k} = \frac{1}{2} \left(\frac{\partial f}{\partial x_k} - i \frac{\partial f}{\partial y_k} \right) \quad \text{and} \quad \frac{\partial f}{\partial \bar{z}_k} = \frac{1}{2} \left(\frac{\partial f}{\partial x_k} + i \frac{\partial f}{\partial y_k} \right)$$

Definition 1.2.2 (Holomorphic Functions of Several Variables)

Let $U \subseteq \mathbb{C}^n$ be open. Let $f = (f_1, \dots, f_n) : U \rightarrow \mathbb{C}$ be a function. Let $p \in U$. We say that f is holomorphic at p if the Cauchy-Riemann condition is satisfied:

$$\frac{\partial f}{\partial \bar{z}_k} = 0$$

for $1 \leq k \leq n$.

Lemma 1.2.3

Let $U \subseteq \mathbb{C}^n$ be open. Let $f = (f_1, \dots, f_n) : U \rightarrow \mathbb{C}$ be a holomorphic function. Then we have

$$\frac{\partial f}{\partial z_k} = \lim_{w \in \mathbb{C} \rightarrow 0} \frac{f(z_1, \dots, z_{k-1}, z_k + w, z_{k+1}, \dots, z_n) - f(z_1, \dots, z_n)}{w}$$

Theorem 1.2.4 (Cauchy's Integral Formula)

Let $U \subseteq \mathbb{C}^n$ be open. Let $f : U \rightarrow \mathbb{C}$ be a holomorphic function. Let $p \in U$. Let $r \in \mathbb{N}^n$. Then we have

$$f(z_1, \dots, z_n) = \frac{1}{(2\pi i)^n} \int_{T_r(p)} \frac{f(w_1, \dots, w_n)}{(w_1 - z_1) \cdots (w_n - z_n)} dw_1 \cdots dw_n$$

for all $(z_1, \dots, z_n) \in D_r(p)$.

Definition 1.2.5 (Analytic Functions)

Let $U \subseteq \mathbb{C}^n$ be open. Let $f : U \rightarrow \mathbb{C}$ be a function. We say that f is analytic if for all $p \in U$, there exists $r \in \mathbb{N}^n$ such that $D_r(p) \subseteq U$ and

$$f(z) = \sum_{\alpha \in \mathbb{N}^n} c_\alpha (z_1 - p_1)^{\alpha_1} \cdots (z_n - p_n)^{\alpha_n}$$

for all $z \in D_r(p)$.

Proposition 1.2.6

Let $U \subseteq \mathbb{C}^n$ be open. Let $f : U \rightarrow \mathbb{C}$ be a continuous function. Suppose that $D_r(p) \subseteq U$ for some $p \in U$ and $r \in \mathbb{N}^n$. Then the function

$$(z_1, \dots, z_n) \mapsto \frac{1}{(2\pi n)!} \int_{T_r(p)} \frac{f(z_1, \dots, z_n)}{(z_1 - p_1) \cdots (z_n - p_n)} dz_1 \cdots dz_n$$

is analytic.

Proposition 1.2.7

Let $U \subseteq \mathbb{C}^n$ be open. Let $f : U \rightarrow \mathbb{C}$ be a function. Then f is holomorphic if and only if it is analytic.

Proposition 1.2.8

Let $U \subseteq \mathbb{C}^n$ be open. Let $f : U \rightarrow \mathbb{C}$ be an analytic function. Suppose that locally at $p = (p_1, \dots, p_n) \in U$, the function is given by

$$f = \sum_{\alpha \in \mathbb{N}^n} c_\alpha (z_1 - p_1)^{\alpha_1} \cdots (z_n - p_n)^{\alpha_n}$$

Then we have

$$c_\alpha = \frac{1}{\alpha_1! \cdots \alpha_n!} \frac{\partial^{\alpha_1}}{\partial z_1^{\alpha_1}} \cdots \frac{\partial^{\alpha_n}}{\partial z_n^{\alpha_n}} f(p)$$

Proposition 1.2.9

Let $U \subseteq \mathbb{C}^n$ be open. Let $f : U \rightarrow \mathbb{C}$ be a holomorphic function. Suppose that $D_r(p) \subseteq U$ for some $r \in \mathbb{N}^n$ and $p = (p_1, \dots, p_n) \in U$. Then we have

$$\frac{\partial^{\alpha_1}}{\partial z_1^{\alpha_1}} \cdots \frac{\partial^{\alpha_n}}{\partial z_n^{\alpha_n}} f(p) = \frac{\alpha_1! \cdots \alpha_n!}{(2\pi i)^n} \int_{T_r(p)} \frac{f(z_1, \dots, z_n)}{(z_1 - p_1)^{\alpha_1+1} \cdots (z_n - p_n)^{\alpha_n+1}} dz_1 \cdots dz_n$$

1.3 Basic Properties of Holomorphic Functions

Theorem 1.3.1 (The Identity Theorem)

Let $U \subseteq \mathbb{C}^n$ be connected and open. Let $f : U \rightarrow \mathbb{C}$ be a holomorphic function. Suppose that $f|_N = 0$ for some open set $\emptyset \subset N \subseteq U$. Then we have $f = 0$.

Theorem 1.3.2 (The Open Mapping Theorem)

Let $U \subseteq \mathbb{C}^n$ be connected and open. Let $f : U \rightarrow \mathbb{C}$ be a non-constant holomorphic function. Then $f(U)$ is connected and open.

Theorem 1.3.3 (The Maximum Principle)

Let $U \subseteq \mathbb{C}^n$ be connected and open. Let $f : U \rightarrow \mathbb{C}$ be a holomorphic function. Suppose that there exist $p \in U$ and an open set $p \in V \subseteq U$ such that $|f(p)| > |f(z)|$ for all $z \in V$. Then we have $f(z) = p$.

The Riemann mapping theorem on $\mathbb{C}^n$ is false for $n \geq 2$.

Example 1.3.4 (Poincare 1907)

$B_1(0) \subseteq \mathbb{C}^2$ and $D_{(1,1)}(0,0)$ are not biholomorphic.

Theorem 1.3.5 (Hartog's Theorem)

Let $n \geq 2$. Let $U \subseteq \mathbb{C}^n$ be open. Let $K \subseteq U$ be compact in U such that $U \setminus K$ is connected. Let $f : U \setminus K \rightarrow \mathbb{C}$ be a holomorphic function. Then f can be extended uniquely to U .

1.4 The Weierstrass Preparation Theorem

Definition 1.4.1 (Weierstrass Polynomials)

A Weierstrass polynomial is a function $W : \mathbb{C}^n \rightarrow \mathbb{C}$ of the form

$$W(z_1, \dots, z_n) = z_n^k + \sum_{j=1}^k a_j(z_1, \dots, z_{n-1}) z_n^{k-j}$$

for some $k \in \mathbb{N}^+$ and $a_j : \mathbb{C}^{n-1} \rightarrow \mathbb{C}$ are holomorphic in $D_{r_j}(0)$ for some $r_j \in \mathbb{N}^+$ and $a_j(0) = 0$.

Theorem 1.4.2 (The Weierstrass Preparation Theorem)

Let $U \subseteq \mathbb{C}^n$ be open such that $0 \in U$. Let $f : U \rightarrow \mathbb{C}$ be a function. Suppose that there exists an open set $V \subseteq U$ such that $0 \in V$ and f is holomorphic on V . Suppose that $f(0, \dots, 0, w)$ is not identically 0. Then there exists a holomorphic function g and a Weierstrass polynomial W such that

$$f = gW$$

and $g \neq 0$ for some neighbourhood $0 \in S \subseteq U$.

Theorem 1.4.3 (The Weierstrass Division Theorem)

2 Domains of Holomorphy

2.1 Basic Definitions

Definition 2.1.1 (Domains of Holomorphy)

Let $\Omega \subseteq \mathbb{C}^n$ be open. We say that Ω is a domain of holomorphy if there are no open sets $U, V \subseteq \mathbb{C}^n$ such that the following are true.

- $U \neq \emptyset$ and $U \subseteq \Omega \cap V$.
- V is connected and $V \not\subseteq \Omega$.
- For any holomorphic function $f : \Omega \rightarrow \mathbb{C}^n$, there exists a holomorphic function $g : V \rightarrow \mathbb{C}$ such that $f|_U = g|_U$.

In other words, Ω is a domain of holomorphy if there exists some holomorphic function on Ω that cannot be extended to a holomorphic function on some larger set than Ω .

Lemma 2.1.2

Let $\Omega \subseteq \mathbb{C}$. Then Ω is a domain of holomorphy.

Lemma 2.1.3

Let $\Omega \subseteq \mathbb{C}^n$ be a convex open set. Then Ω is a domain of holomorphy.

2.2 Different Notions of Convexity in $\mathbb{C}^n$

Definition 2.2.1 (Holomorphic Convexity)

Definition 2.2.2 (Pseudoconvexity)

2.3 Plurisubharmonic Functions

2.4 Equivalent Characterizations of Domains of Holomorphy

Proposition 2.4.1

Let $\Omega \subseteq \mathbb{C}^n$ be open. Then the following are equivalent.

- Ω is a domain of holomorphy.
- Ω is holomorphically convex.
- Ω is pseudoconvex.

2.5 The Levi Problem

3 Solutions to the $\bar{\partial}$ Equation